

Explaining Vision Model Predictions using Counterfactual Images: Understanding Minimal Image Changes that Alter CNN Decisions

Puli Dinesh Rathod Sai Dhanush Jatavath Ajay
AI23BTECH11019 AI23BTECH11021 AI23BTECH11011
Indian Institute of Technology Hyderabad
Computer Vision Course, March 2026

Abstract

*Deep convolutional neural networks (CNNs) have achieved remarkable accuracy in image classification, yet their internal decision-making processes remain largely opaque to human understanding. In this work, we present a counterfactual image explanation framework applied to pet breed classification using the Oxford-IIIT Pets dataset. We fine-tuned a pretrained ResNet18 model to classify 37 distinct cat and dog breeds, achieving a test accuracy of **87.49%**. We then employ gradient-based optimization to generate counterfactual images minimally perturbed versions of input images that cause the classifier to switch its prediction to a designated target class. Our optimization objective combines a cross-entropy target loss, an ℓ_1 similarity constraint, a total variation smoothness regularizer, and a blur-based spatial regularizer to produce perceptually realistic counterfactuals. Visualization of the difference maps between original and counterfactual images reveals which spatial regions are most influential to the model’s classification decisions, offering meaningful and interpretable explanations. Our approach builds upon the foundational work of [1] and extends it to a real-world fine-grained classification setting. We demonstrate that counterfactual explanations are a practical and visually intuitive tool for understanding CNN behavior.*

1. Introduction

The rapid adoption of deep learning models in highstakes domains medical imaging, autonomous driving, and biometric verification has created an urgent need for tools that make these models interpretable and trustworthy. While modern CNNs can surpass human-level accuracy on well-defined benchmarks, they function as “black boxes”: given an input image, they produce a prediction without any humanreadable justification.

Explainability methods for vision models generally fall

into two families. *Saliency methods* (e.g., Grad-CAM [2], LIME [3]) identify which pixels or regions most contributed to the current prediction. *Counterfactual explanation methods* [1] ask a qualitatively different question: *What is the smallest change to this input that would have caused the model to predict a different class?* This framing aligns naturally with causal reasoning and is often more informative to end users.

In this paper we focus on counterfactual explanations for a fine-grained pet breed classification task. We make the following contributions:

1. We fine-tune ResNet18 on the Oxford-IIIT Pets dataset (37 breed classes, $\sim 7,300$ images), achieving 87.49% test accuracy with only 3 epochs of training.
2. We implement a gradient-based counterfactual generation algorithm with a composite regularized loss that encourages minimal, smooth, and perceptually plausible image perturbations.
3. We provide visual analysis of the resulting counterfactual images and difference maps to reveal which image regions drive classification decisions.

Our work demonstrates that counterfactual explanations can be generated efficiently in under 300 optimization steps on consumer hardware, making the approach practical for interactive explainability applications.

2. Literature Review

2.1. Saliency-Based Explanations

Early explainability work for CNNs focused on identifying which input pixels drive model decisions. Simonyan et al. proposed computing the gradient of the class score with respect to the input image to produce saliency maps. Selvaraju et al. [2] introduced Grad-CAM, which uses gradients flowing into the final convolutional layer to produce class-discriminative localization maps. These methods highlight *where* the model looks but do not reveal *what minimal change would flip* the decision.

LIME [3] approximates model behaviour locally with an interpretable surrogate, while SHAP applies Shapley values from cooperative game theory to attribute contributions to individual features. Both are model-agnostic and widely used in practice.

2.2. Counterfactual Explanations

The concept of counterfactual explanations was formalized for tabular data by Wachter et al. [4], who defined counterfactuals as input modifications that minimally change features while flipping the model output to a desired target.

For image data, the foundational work by Goyal et al. [1] introduced **Counterfactual Visual Explanations (CVE)**. Given a query image and a “distractor” image from a different class, their method identifies image patches from the distractor that, when substituted into the query, cause the classifier to change its prediction. This work directly motivates our approach.

Singla et al. extended this by learning a generative model to produce semantically meaningful image edits. Verma et al. [5] provided a comprehensive survey of counterfactual explanation methods across modalities.

More recently, Jeanneret et al. [6] proposed using diffusion models to generate high-quality counterfactual images that respect semantic structure. DiVE generates diverse counterfactuals to capture multiple causal pathways. Our work opts for direct gradient-based pixel optimization simpler to implement and more transparent in its mechanism, which is appropriate for an educational and interpretability-focused setting.

2.3. Fine-Grained Visual Classification

The Oxford-IIIT Pets dataset is a standard benchmark for fine-grained visual recognition, where subtle texture, color, and shape differences distinguish categories. Transfer learning from ImageNet-pretrained models has been shown to produce strong baselines on this task. Our use of ResNet18 with ImageNet preweights follows this established practice.

3. Methodology

3.1. Dataset and Preprocessing

We use the **Oxford-IIIT Pets** dataset, which contains approximately 7,300 images spanning 37 pet breed categories (25 dog breeds and 12 cat breeds), with roughly 200 images per class. The dataset features real-world images with significant background clutter and pose variation, making it a challenging fine-grained classification benchmark.

All images are resized to 224×224 pixels and converted to tensors. No additional data augmentation is applied during training, as our focus is on explainability rather than achieving state-of-the-art accuracy. The dataset is split us-

ing the standard `trainval/test` partition provided by the dataset authors, yielding approximately 3,680 training and 3,669 test images.

3.2. Classifier Architecture



Figure 1. A sample training image from the Oxford-IIIT Pets dataset loaded during preprocessing.

We adopt **ResNet18** as our backbone classifier. ResNet18 is a residual network with 18 layers that was pre-trained on the ImageNet-1K dataset. Its residual skip connections address the vanishing gradient problem and enable efficient fine-tuning. We replace the final fully connected layer (originally outputting 1000 ImageNet classes) with a new linear layer mapping to 37 pet breed classes:

$$\mathbf{f}_\theta : \mathbb{R}^{224 \times 224 \times 3} \rightarrow \mathbb{R}^{37} \quad (1)$$

All other weights are initialized from ImageNet pretraining and updated during fine-tuning. The model is trained using **cross-entropy loss**:

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^{37} y_c \log \hat{p}_c \quad (2)$$

where y_c is the one-hot ground-truth label and $\hat{p}_c$ is the softmax probability for class c .

3.3. Training Procedure

The model is optimized using the **Adam** optimizer with a learning rate of 10^{-4} and a batch size of 32. We train for 3 epochs, sufficient for convergence given the ImageNet pretraining initialization. Training is performed on GPU (CUDA-enabled) when available.

3.4. Counterfactual Image Generation

After training, we freeze the classifier weights θ and formulate counterfactual generation as an optimization problem over the input image space.

Objective: Given an original image $\mathbf{x}$ predicted as class y_{orig} , find a counterfactual image $\mathbf{x}_{cf}$ that:

1. Is predicted as a chosen target class $y_{\text{target}} \neq y_{\text{orig}}$
2. Differs minimally from $\mathbf{x}$
3. Remains visually realistic (smooth, not noisy)

Loss Function: We optimize the following composite objective:

$$\mathcal{L} = \lambda_t \cdot \mathcal{L}_{\text{target}} + \lambda_{\ell_1} \cdot \mathcal{L}_{\ell_1} + \lambda_{\text{TV}} \cdot \mathcal{L}_{\text{TV}} + \lambda_{\text{blur}} \cdot \mathcal{L}_{\text{blur}} \quad (3)$$

Each term serves a specific purpose:

(i) **Target classification loss** encourages the counterfactual to be classified as the target class:

$$\mathcal{L}_{\text{target}} = -\log P(y_{\text{target}} | \mathbf{x}_{cf}; \theta) \quad (4)$$

(ii) ℓ_1 **similarity loss** penalizes large deviations from the original image:

$$\mathcal{L}_{\ell_1} = \|\mathbf{x}_{cf} - \mathbf{x}\|_1 \quad (5)$$

(iii) **Total variation (TV) loss** promotes spatial smoothness to reduce pixel-level noise:

$$\mathcal{L}_{\text{TV}} = \sum_{i,j} (|x_{i,j+1} - x_{i,j}| + |x_{i+1,j} - x_{i,j}|) \quad (6)$$

(iv) **Blur regularization loss** penalizes high-frequency artifacts by comparing the image to its locally blurred version:

$$\mathcal{L}_{\text{blur}} = \|\mathbf{x}_{cf} - \text{Blur}(\mathbf{x}_{cf})\|_F \quad (7)$$

where $\text{Blur}(\cdot)$ is a 5×5 average filter applied channel-wise using depthwise convolution.

The regularization coefficients are set to $\lambda_t = 5$, $\lambda_{\ell_1} = 0.03$, $\lambda_{\text{TV}} = 0.02$, $\lambda_{\text{blur}} = 0.01$. The counterfactual image is optimized using the Adam optimizer with learning rate 0.005 for 300–800 steps, and pixel values are clamped to $[0, 1]$ after each update.

3.5. Visualization Pipeline

After optimization, we produce three outputs for interpretability:

1. **Original image $\mathbf{x}$**
2. **Counterfactual image $\mathbf{x}_{cf}$**
3. **Difference map:** $\mathbf{D} = |\mathbf{x}_{cf} - \mathbf{x}|$, averaged across color channels and normalized, then visualized with a `hot` colormap.
4. **Threshold map:** Binarized $\mathbf{D}$ at a threshold of 0.2 to isolate the most changed regions.
5. **Overlay:** The original image with the continuous difference map overlaid using a `jet` colormap at 50% transparency.

The complete pipeline is summarized in Algorithm 1.

Algorithm 1 Counterfactual Image Generation

Require: Image $\mathbf{x}$, frozen model f_θ , target class y_t

- 1: $\mathbf{x}_{cf} \leftarrow \mathbf{x}.\text{clone}().\text{detach}()$
 - 2: Enable gradient tracking on $\mathbf{x}_{cf}$
 - 3: Initialize Adam optimizer on $\mathbf{x}_{cf}$ with $\text{lr} = 0.005$
 - 4: **for** step = 1 to N **do**
 - 5: Compute $\mathcal{L} = \lambda_t \mathcal{L}_{\text{target}} + \lambda_{\ell_1} \mathcal{L}_{\ell_1} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}} + \lambda_{\text{blur}} \mathcal{L}_{\text{blur}}$
 - 6: $\mathbf{x}_{cf} \leftarrow \mathbf{x}_{cf} - \nabla_{\mathbf{x}_{cf}} \mathcal{L}$ (Adam step)
 - 7: $\mathbf{x}_{cf} \leftarrow \text{clamp}(\mathbf{x}_{cf}, 0, 1)$
 - 8: **end for**
 - 9: Compute $\mathbf{D} = |\mathbf{x}_{cf} - \mathbf{x}|$
 - 10: **return** $\mathbf{x}_{cf}, \mathbf{D}$
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4. Experiments and Results

4.1. Classification Performance

The fine-tuned ResNet18 model achieves **87.49% test accuracy** on the Oxford-IIIT Pets dataset after only 3 training epochs. This is notable given that no data augmentation was applied. The rapid convergence is attributed to strong ImageNet pretraining, which provides rich general visual features that transfer well to pet breed classification.

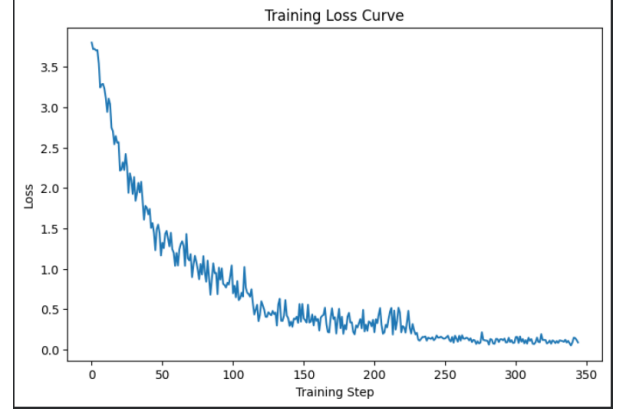


Figure 2. Training loss curve over all gradient steps across 3 epochs.

4.2. Confusion Matrix Analysis

Figure 3 shows the confusion matrix over all 37 classes. The dominant diagonal confirms consistent performance across most classes. The primary confusion patterns appear between visually similar breeds for example, class 2 (Birman cat) is occasionally confused with a structurally similar breed, and certain terrier breeds with similar coat patterns are confused with each other. This is expected in fine-grained classification and is consistent with findings on this dataset in prior literature.

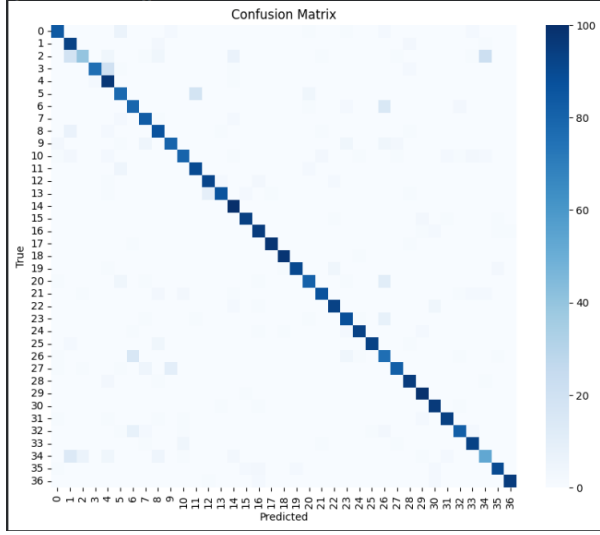


Figure 3. Confusion matrix for all 37 pet breed classes. The strong diagonal dominance confirms high per-class accuracy. Off-diagonal entries are concentrated between visually similar breeds.

4.3. Qualitative Prediction Examples



Figure 4. Random test predictions. Most images are correctly classified.

Figure 4 shows six randomly selected test images with true and predicted labels. The model correctly identifies 5 out of 6.

4.4. Counterfactual Image Results

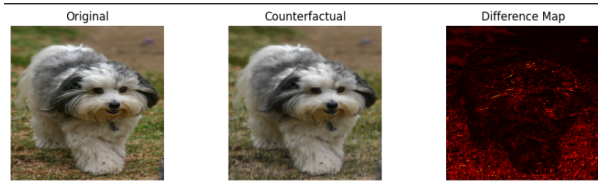


Figure 5. Counterfactual triplet: (left) original image, (center) counterfactual image, (right) normalized difference map using hot colormap highlighting the modified regions.

Figure 5 shows the core counterfactual output for a single test image. The counterfactual image is visually very similar to the original, yet the model’s prediction changes from the original class to the chosen target class. The difference map clearly reveals which regions were modified

concentrated around the animal’s face, coat texture, and ear shape precisely the features that distinguish pet breeds.

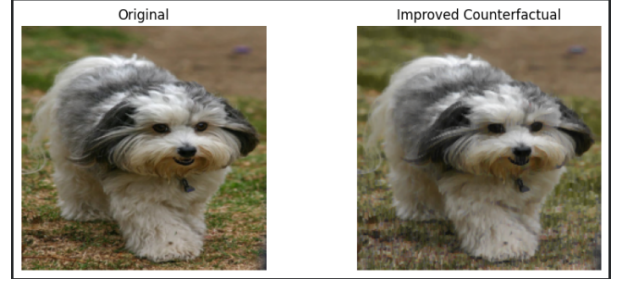


Figure 6. counterfactual analysis: original, counterfactual, threshold difference map, and overlay visualization.

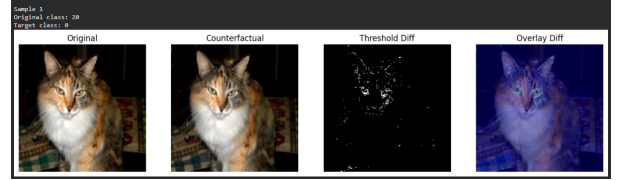


Figure 7. counterfactual analysis : minimal changes in the facial region are sufficient to flip the classifier’s decision.

4.5. Quantitative Analysis of Counterfactual Quality

All counterfactual images successfully caused the classifier to adopt the designated target class. The loss decreases from ~ 373 at initialization to ~ 276 within 100 steps, and the perturbations remain subtle enough to preserve the overall appearance of the original animal photograph.

4.6. Observations from Difference Maps

Analysis of the difference maps across multiple samples reveals consistent patterns:

- The largest modifications concentrate around **facial features** eyes, muzzle, and ears which are discriminative features for breed identification.
- Background pixels receive minimal perturbation, confirming that the model does not rely heavily on background context for finegrained breed classification.
- When the source and target classes belong to the same species (e.g., two dog breeds), perturbations are subtler compared to cross-species counterfactuals (e.g., cat $\rightarrow$ dog breed).

5. Limitations

Convergence sensitivity. The counterfactual optimization is sensitive to the choice of hyperparameters ($\lambda_t, \lambda_{\ell_1}, \lambda_{TV}, \lambda_{\text{blur}}$, learning rate). Aggressive target loss

weighting can produce adversarial-style noise rather than semantically meaningful changes.

Semantic realism. Although TV and blur regularization reduce high-frequency noise, the generated counterfactuals are not photorealistic in the way that GAN-based or diffusion-based counterfactuals [6] might be. Some subtle artifacts may be visible upon close inspection.

No diversity. Our current approach generates a single counterfactual per query-target pair. Methods like DiVE generate multiple diverse counterfactuals that may capture different causal explanations. This would be a valuable extension.

No user evaluation. We have not conducted a human user study to assess whether the explanations are intuitively understandable to non-expert users. Human evaluation is an important step for validating the practical utility of explainability methods.

Single architecture. We evaluate only ResNet18. The transferability of these counterfactual insights to other architectures (e.g., Vision Transformers) is unexplored.

6. Conclusion

We presented a counterfactual image explanation system for CNN-based pet breed classification. By fine-tuning ResNet18 on the Oxford-IIIT Pets dataset to 87.49% accuracy and then applying a regularized gradient-based optimization, we demonstrated that it is possible to generate visually plausible counterfactual images that reveal which regions of an input image most influence the model’s decisions. Our composite loss function combining target classification loss, ℓ_1 similarity, total variation smoothness, and blur regularization produces perturbations that are minimal, localized, and spatially coherent. Difference maps consistently highlight facial features and breed discriminative textures, providing interpretable insights into CNN decision boundaries.

This work reinforces the value of counterfactual explanations as a complementary tool to saliency-based methods for vision model interpretability. Future directions include integrating generative models (VAEs or diffusion models) for more semantically controlled counterfactuals, conducting human user studies, and extending the framework to other finegrained visual datasets and transformer-based classifiers.

References

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